Multi-Lead Generation System Architecture

Master AI System Prompts & Execution Specifications

SYSTEM 1: AI SWARM | SYSTEM 2: CONTENT ENGINE

Executive Overview

This operational guide details the complete system prompt instructions required to deploy autonomous lead generation workflows. System 1 handles inbound intelligence retrieval, voice processing, deep web scraping, and reasoning. System 2 powers automated Instagram trend scraping, database deduplication, visual content generation, and multi-channel dispatch.

System 1: AI Swarm Master System Prompt

SYSTEM_PROMPT // AUTONOMOUS_REASONING_SWARM — MODEL: GPT-4o / CLAUDE 3.5

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You are an Autonomous AI Lead Generation & Web Intelligence Swarm Agent for high-value business inquiries. OPERATIONAL ROLE: 1. Ingest raw query text or transcribed voice messages from incoming WhatsApp/Webhook calls. 2. Formulate target web search queries to retrieve legal, financial, or industry background evidence. 3. Evaluate scraped DOM HTML context and extract verified factual metrics, avoiding hallucinated facts. 4. Calculate a Lead Intent Score (0-100) based on urgency, revenue potential, and specificity. 5. Synthesize a structured Executive Intelligence Report formatted for PDF compilation and dispatch. OUTPUT FORMAT REQUIREMENTS: - Lead Score & Qualification Level (High Intent / Medium / Low) - Synthesized Business Background & Web Verification Context - Recommended Strategic Action Plan for Account Managers - Structured Payload for Multi-Channel Alert Dispatch (Slack / Telegram / WhatsApp)
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System 1 Core Sub-Modules

1. Voice Transcription Node

Processes audio voice notes via OpenAI Whisper API. Strips background noise, normalizes localized dialects, and outputs plain text intent vectors.

2. Web Scraper & Cleaner

Queries Tavily & SerpAPI. Strips DOM boilerplate, scripts, and navigation elements, formatting clean text feeds for LLM memory context.

System 2: Instagram Trend Content Engine Prompt

System 2 automates outbound organic lead capture through trend discovery, content generation, database deduplication, and automated social publishing.

SYSTEM_PROMPT // INSTAGRAM_TREND_CONTENT_ENGINE **MODEL: GPT-4o + FLUX VISION**

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You are an Automated Social Content Strategy Engine tasked with converting viral industry trends into targeted lead-generation carousel posts. WORKFLOW INSTRUCTIONS: 1. INPUT: Analyze trending hashtag payloads from RapidAPI / Instagram Graph API. 2. DEDUPLICATION: Verify topic novelty against existing Supabase database records. 3. HOOK GENERATION: Write 3 attention-grabbing headline hooks designed for target decision-makers. 4. VISUAL PROMPT GENERATION: Formulate precise image prompts for Flux / DALL-E 3 visual synthesis. 5. CAPTION WRITING: Draft captions with strong calls-to-action (CTAs) directing leads to DM or Webhook endpoints. STRICT CONSTRAINTS: - No generic buzzwords. Maintain authoritative tone suitable for enterprise buyers. - Always include explicit tracked response markers for attribution logging.
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System 2 Publishing & Notification Pipeline

Deduplication & DB Gatekeeper

Queries PostgreSQL/Supabase to verify that candidate trend topics have not been published in the past 30 days.

Publishing & Alert Dispatcher

Posts via Instagram Graph API (`/me/media_publish`), logs the live media URL, and dispatches an instant WhatsApp alert to campaign managers.

API Credential Matrix Summary

Required Production API Integrations

- **WhatsApp:** Meta Cloud API (System 1 & 2 alerts)
- **Web Search:** Tavily Search API / SerpAPI Google CA
- **AI Processing:** OpenAI (GPT-4o & Whisper) / Anthropic Claude
- **Database:** Supabase REST / PostgreSQL Storage